

Q1. Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.

Given Relation

Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price)

Functional Dependencies

The functional dependencies are:

1. OrderID $\rightarrow$ CustID
2. CustID $\rightarrow$ CustName
3. ProdID $\rightarrow$ ProdName, Price
4. (OrderID, ProdID) $\rightarrow$ Qty

Since an order can contain multiple products, the candidate key is:

Candidate Key = (OrderID, ProdID)

Conversion to 1NF

The relation is already in First Normal Form (1NF) because all attributes contain atomic values and there are no repeating groups.

Conversion to 2NF

The relation is not in 2NF because partial dependencies exist:

- OrderID $\rightarrow$ CustID
- ProdID $\rightarrow$ ProdName, Price

These attributes depend only on part of the composite key (OrderID, ProdID).

Decompose the relation into:

Customer

Customer(CustID, CustName)

Product

Product(ProdID, ProdName, Price)

Order

Order(OrderID, CustID)

OrderItem

OrderItem(OrderID, ProdID, Qty)

Conversion to 3NF

There is a transitive dependency:

$\text{OrderID} \rightarrow \text{CustID} \rightarrow \text{CustName}$

This is removed by keeping customer information in a separate relation.

Final 3NF Relations

- Customer(CustID, CustName)
- Product(ProdID, ProdName, Price)
- Order(OrderID, CustID)
- OrderItem(OrderID, ProdID, Qty)

Thus, the relation is normalized to 3NF, eliminating insertion, deletion, and update anomalies.

Q2. A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.

Given Relation

Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName)

Functional Dependencies

1. $\text{PID} \rightarrow \text{PName}, \text{DocID}, \text{WardNo}$
2. $\text{DocID} \rightarrow \text{DocName}, \text{Specialization}$
3. $\text{WardNo} \rightarrow \text{WardName}$

The candidate key is:

Candidate Key = PID

Anomalies

1. **Insertion Anomaly:**

A doctor cannot be added to the database unless a patient is associated with that doctor.

2. Update Anomaly:

If a doctor's name or specialization changes, the information must be updated in multiple patient records.

3. Deletion Anomaly:

If the last patient associated with a doctor is deleted, the doctor's information may also be lost.

BCNF Decomposition

Patient Relation

Patient(PID, PName, DocID, WardNo)

Here:

$PID \rightarrow PName, DocID, WardNo$

Since PID is the candidate key, this relation satisfies BCNF.

Doctor Relation

Doctor(DocID, DocName, Specialization)

Here:

$DocID \rightarrow DocName, Specialization$

DocID is the candidate key.

Ward Relation

Ward(WardNo, WardName)

Here:

$WardNo \rightarrow WardName$

WardNo is the candidate key.

Final BCNF Relations

- Patient(PID, PName, DocID, WardNo)
- Doctor(DocID, DocName, Specialization)
- Ward(WardNo, WardName)

Therefore, the database is normalized to BCNF.

Q3. University stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.

Functional Dependencies

1. $SID \rightarrow SName$
2. $CourseID \rightarrow CourseName, Faculty$
3. $(SID, CourseID) \rightarrow Grade$

Candidate Key

A student can enroll in several courses and a course can have several students.

Therefore:

Candidate Key = (SID, CourseID)

1NF

The relation is in 1NF because all attributes contain atomic values.

Partial Dependencies

The following partial dependencies exist:

$SID \rightarrow SName$

$CourseID \rightarrow CourseName, Faculty$

These attributes depend only on part of the composite key.

2NF Decomposition

Student

Student(SID, SName)

Course

Course(CourseID, CourseName, Faculty)

Enrollment

Enrollment(SID, CourseID, Grade)

In the decomposed relations, every non-key attribute depends on the complete primary key.
Therefore, the final design is in 2NF.

Q4. Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.

Functional Dependencies

1. $\text{PassID} \rightarrow \text{PassName}$
2. $\text{FlightNo} \rightarrow \text{DeptCity, ArrCity, Date}$
3. $(\text{FlightNo, PassID}) \rightarrow \text{Seat}$

Candidate Key

Assuming one passenger can make one booking for a particular flight:

Candidate Key = (FlightNo, PassID)

1NF

The relation is already in 1NF because all attributes are atomic.

2NF

Partial dependencies exist:

$\text{PassID} \rightarrow \text{PassName}$

$\text{FlightNo} \rightarrow \text{DeptCity, ArrCity, Date}$

Therefore, decompose the relation.

3NF

Separate passenger and flight information.

Passenger(PassID, PassName)

Flight(FlightNo, DeptCity, ArrCity, Date)

Booking(FlightNo, PassID, Seat)

BCNF

In each relation, every determinant is a candidate key:

PassID is the key of Passenger.

FlightNo is the key of Flight.

(FlightNo, PassID) is the key of Booking.

Therefore, the final relations satisfy BCNF.

Q5. Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.

Consider the relation:

Employee(EmpID, EmpName, DeptID, DeptName, ManagerID, ManagerName, JobID, JobTitle)

Functional Dependencies

1. $\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}, \text{ManagerID}, \text{JobID}$
2. $\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$
3. $\text{ManagerID} \rightarrow \text{ManagerName}$
4. $\text{JobID} \rightarrow \text{JobTitle}$

Transitive Dependencies

The following transitive dependencies exist:

$\text{EmpID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}$

Therefore:

$\text{EmpID} \rightarrow \text{DeptName}$

Similarly:

$\text{EmpID} \rightarrow \text{ManagerID} \rightarrow \text{ManagerName}$

Therefore:

$\text{EmpID} \rightarrow \text{ManagerName}$

Also:

$\text{EmpID} \rightarrow \text{JobID} \rightarrow \text{JobTitle}$

Therefore:

$\text{EmpID} \rightarrow \text{JobTitle}$

3NF Decomposition

Employee

Employee(EmpID, EmpName, DeptID, ManagerID, JobID)

Department

Department(DeptID, DeptName, ManagerID)

Manager

Manager(ManagerID, ManagerName)

Job

Job(JobID, JobTitle)

Result

The transitive dependencies have been removed. Every non-key attribute now depends directly on the key of its relation.

Therefore, the schema is in 3NF.

Q6. Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.

Functional Dependencies

$\text{MemberID} \rightarrow \text{MemberName}$

$\text{BookID} \rightarrow \text{Title, Genre}$

Multivalued Dependencies

A member may borrow multiple books:

$\text{MemberID} \twoheadrightarrow \text{BookID}$

A book may have multiple authors:

BookID $\twoheadrightarrow$ Author

These independent multivalued dependencies cause redundancy.

4NF Decomposition

Member

Member(MemberID, MemberName)

Book

Book(BookID, Title, Genre)

BookAuthor

BookAuthor(BookID, Author)

Borrow

Borrow(MemberID, BookID, BorrowDate)

The independent multivalued facts are separated into different relations.

Therefore, the database is normalized to 4NF.

Q7. Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.

Consider:

Billing(CustomerID, CustomerName, PhoneNo, PlanID, PlanName, CallID, CallDuration, Rate, BillID, BillAmount)

Functional Dependencies

1. CustomerID $\rightarrow$ CustomerName
2. PhoneNo $\rightarrow$ CustomerID, PlanID
3. PlanID $\rightarrow$ PlanName, Rate
4. CallID $\rightarrow$ PhoneNo, CallDuration
5. BillID $\rightarrow$ CustomerID, BillAmount

Candidate Keys

Based on the assumed uniqueness constraints:

- CustomerID
- PhoneNo
- CallID
- BillID

These can act as candidate keys where each uniquely identifies its corresponding entity/record.

Primary Key

A suitable primary key for the customer entity is:

CustomerID

Transitive Dependency

PhoneNo $\rightarrow$ PlanID

and

PlanID $\rightarrow$ PlanName, Rate

Therefore:

PhoneNo $\rightarrow$ PlanName, Rate

This is a transitive dependency that should be removed during normalization.

Q8. A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.

Functional Dependencies

1. PID $\rightarrow$ PName, SupplierID, Category, Price, Warehouse

2. SupplierID $\rightarrow$ SupplierName

Therefore:

PID $\rightarrow$ SupplierID $\rightarrow$ SupplierName

This is a transitive dependency.

1NF

All attributes are atomic, so the relation is in 1NF.

2NF

The candidate key is PID, which is a single attribute. Therefore, there can be no partial dependency.

Hence, it is in 2NF.

3NF

Remove the transitive dependency by decomposing the relation.

Product

Product(PID, PName, SupplierID, Category, Price, Warehouse)

Supplier

Supplier(SupplierID, SupplierName)

Now supplier information is stored separately.

Final Design

Product(PID, PName, SupplierID, Category, Price, Warehouse)

Supplier(SupplierID, SupplierName)

The database is normalized to 3NF.

Q9. Study the poorly normalized School ERP schema and propose a fully normalized design up to BCNF with justification.

Consider:

School(StudentID, StudentName, ClassID, ClassName, TeacherID, TeacherName, SubjectID, SubjectName, Marks)

Functional Dependencies

1. StudentID $\rightarrow$ StudentName, ClassID
2. ClassID $\rightarrow$ ClassName, TeacherID
3. TeacherID $\rightarrow$ TeacherName

4. $\text{SubjectID} \rightarrow \text{SubjectName}$

5. $(\text{StudentID}, \text{SubjectID}) \rightarrow \text{Marks}$

Candidate Key

Since a student can study multiple subjects:

Candidate Key = (StudentID, SubjectID)

Partial Dependencies

$\text{StudentID} \rightarrow \text{StudentName}, \text{ClassID}$

$\text{SubjectID} \rightarrow \text{SubjectName}$

These depend on only part of the composite key.

Transitive Dependencies

$\text{StudentID} \rightarrow \text{ClassID} \rightarrow \text{ClassName}$

$\text{StudentID} \rightarrow \text{ClassID} \rightarrow \text{TeacherID}$

$\text{TeacherID} \rightarrow \text{TeacherName}$

BCNF Decomposition

Student

Student(StudentID, StudentName, ClassID)

Class

Class(ClassID, ClassName, TeacherID)

Teacher

Teacher(TeacherID, TeacherName)

Subject

Subject(SubjectID, SubjectName)

StudentSubject

StudentSubject(StudentID, SubjectID, Marks)

Justification

Each determinant is a candidate key in its respective relation:

- $\text{StudentID} \rightarrow \text{Student}$
- $\text{ClassID} \rightarrow \text{Class}$
- $\text{TeacherID} \rightarrow \text{Teacher}$
- $\text{SubjectID} \rightarrow \text{Subject}$
- $(\text{StudentID}, \text{SubjectID}) \rightarrow \text{StudentSubject}$

Hence, the resulting relations satisfy BCNF and remove redundancy and anomalies.

Q10. Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.

Consider:

$\text{Streaming}(\text{MovieID}, \text{ActorID}, \text{DirectorID}, \text{PlatformID})$

Assume that actors, directors, and streaming platforms can be associated independently with a movie.

Join Dependencies

The relation contains three independent relationships:

1. Movie–Actor
2. Movie–Director
3. Movie–Platform

Therefore, the join dependency can be represented as:

$\text{Streaming} = \text{MovieActor} \bowtie \text{MovieDirector} \bowtie \text{MoviePlatform}$

5NF Decomposition

MovieActor

$\text{MovieActor}(\text{MovieID}, \text{ActorID})$

MovieDirector

$\text{MovieDirector}(\text{MovieID}, \text{DirectorID})$

MoviePlatform

MoviePlatform(MovieID, PlatformID)

Lossless-Join Verification

The original relation can be reconstructed using:

MovieActor $\bowtie$ MovieDirector $\bowtie$ MoviePlatform

If the join produces exactly the valid tuples of the original relation, the decomposition is lossless.

Final 5NF Design

MovieActor(MovieID, ActorID)

MovieDirector(MovieID, DirectorID)

MoviePlatform(MovieID, PlatformID)

Thus, the decomposition removes redundancy caused by join dependencies and results in a 5NF design.